

AinPath — Exam Preparation Module

MCQ Question Bank · Model Tests · Student Accounts — বার কাউন্সিল এনরোলমেন্ট ও বিজেএস (সহকারী জজ) পরীক্ষার প্রস্তুতি · Implementation Plan · 2026-09-04

Context & Confirmed Decisions

AinPath already models Bar Council Enrolment and BJS as exam programs (`has_exam_stages = true`) with seeded subjects, but is PDF-content-only. This module makes preparation interactive. Confirmed with the product owner:

- **Student accounts** (register/login) — a new auth layer; the public site is currently 100% anonymous.
- **All four v1 features:** subject-wise MCQ practice (instant answer + explanation), timed model tests (timer, negative marking, result + review), previous-year question practice/browse, written question archive.
- **Question entry:** admin form + CSV bulk import (native `fgetcsv`, no new composer package).
- **Access split:** archive browsing stays public/anonymous; interactive practice and model tests require student login.
- Students live in a separate `students` table — never in the staff `users` table / Spatie roles.

All work follows repo conventions: Route → Controller → FormRequest → Service (final, transactions on multi-write) → Model (`#[Fillable]`, `casts()`, `HasTranslatedFields`) → Resource; ApiResponse { `success`, `message`, `result` }; `simplePaginate`; Spatie route-middleware permissions from `config/admin-permissions.php`; render-only Inertia page controllers; TypeScript frontend with the `@/components/ui` barrel, `useResourceIndex / DataTable` (admin) and `usePublicList / usePublicResource` (public); single-file i18n (`resources/js/lib/i18n.ts`, `bn = source of truth`); no comments in new code.

Part A — Backend

A1. Student auth

Dedicated student session guard (driver `session`, provider `students`) — not sanctum-driver, not statefulApi. Sanctum's session path doesn't provider-check, so putting students on `sanctum.guard` would leak student sessions into `auth:sanctum admin` endpoints. Separate guards keep audiences isolated; a staff member can simultaneously be a logged-in student (separate `login_student_*` session key). Student logout = `Auth::guard('student')->logout() + session()->regenerate()` — never `invalidate()`.

- `config/auth.php`: `student` guard, `students` eloquent provider, `students` password broker with its own `student_password_reset_tokens` table.
- `bootstrap/app.php`: fourth mount — `web` middleware, prefix `v1/student`, name `v1.student`, file `routes/student-v1.php`. Replace `redirectGuestsTo('/admin/login')` with a closure: admin paths → `/admin/login`, else `/locale/account/login?redirect={path}`.
- `AppServiceProvider`: `RateLimiter::for('student-auth', 10/min by IP)`.
- New middleware `EnsureStudentIsActive` (403 on deactivated account, mid-session).
- `Student` gets `HasApiTokens` now; a `student-api` sanctum guard for mobile is future work, zero schema change.

A2. Migrations (after 2026_09_05_000008)

File	Tables
<code>...000009_create_students_tables</code>	students (name, email unique, phone, password, program_id FK nullable, email_verified_at, is_active, last_login_at, rememberToken); student_password_reset_tokens (email pk, token, created_at)
<code>...000010_create_question_bank_tables</code>	questions (type mcq written, subject_id FK restrict, exam_stage, exam_year, question_bn/en, explanation_bn/en, reference, status draft/published, created_by/updated_by, softDeletes; composite indexes); question_options (question_id cascade, option_bn/en, is_correct, sort_order)
<code>...000011_create_model_test_tables</code>	model_tests (slug unique, title_bn/en, description_bn/en, program_id FK, exam_stage, duration_minutes, negative_mark decimal(4,2) default 0.25, status, published_at, softDeletes); model_test_questions (pivot: sort_order, marks decimal default 1, unique [test, question])

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...000012_create_test_attempt_tables
```

test_attempts (student_id cascade, model_test_id restrict, status in_progress/submitted/expired, **active** bool **NULLABLE** — unique [student, test, active] gives a DB-level one-in-progress-attempt guard since NULLs don't collide; started/expires/submitted_at, score decimal(6,2), correct/wrong/skipped counts); **attempt_answers** (attempt cascade, question, chosen option nullable, is_correct snapshot, unique [attempt, question]); **practice_sessions** (student, subject, question_count, correct_count) — v1 practice history

A3. Enums & models

New enums `QuestionType` (mcq/written), `AttemptStatus` (in_progress/submitted/expired) with bilingual `labels()`; reuse `ContentStatus` and `ExamStage`. Models: `Student` (extends Foundation Auth User; HasApiTokens, Notifiable; hashed password; `StudentResetPassword` notification cloned from `AdminResetPassword`), `Question`, `QuestionOption`, `ModelTest` (questions() withPivot ordered, totalMarks, isPubliclyVisible), `TestAttempt` (isPastExpiry, remainingSeconds), `AttemptAnswer`, `PracticeSession`.

A4. Routes

- **routes/student-v1.php** (new; controllers in `v1\StudentApi\`): auth register/login (throttle student-auth), forgot/reset-password; under `auth:student` + `EnsureStudentsIsActive`: me, logout, profile; practice subjects/questions/sessions; model-tests index/show; POST `model-tests/{slug}/attempts` (start/resume); attempts index, GET `attempts/{id}` (rehydrate: questions *without* answers + saved answers + remaining_seconds), PUT answers, POST submit, GET result. Ownership via `abort_unless(..., 404)`.
- **routes/public-v1.php**: `question-archive/filters|mcq|written`; anonymous `model-tests` catalogue (no question payload).
- **routes/admin-v1.php**: questions filters/import-template/import/publish/unpublish + `apiResource`; model-tests filters/publish/unpublish + questions attach/reorder/detach + `apiResource`; students index/show/toggle-active. Builder picker reuses `GET /v1/admin/questions`.
- **routes/web.php** (render-only pages, locale group): public `/exam-prep`, `/questions`, `/model-tests/{slug}`; `guest:student` account login/register/forgot/reset; `auth:student` practice, practice/run, runner, profile, attempts. **routes/web-admin.php**: Question/ModelTest/Student page controllers (id-only props).

A5. Service-layer core logic

- **QuestionService** — CRUD in transactions with option sync; publish guard: mcq must have exactly one correct option.
- **QuestionImportService** — CSV header `subject,type,exam_stage,exam_year,question_bn,question_en,option_1..5,correct_option,explanation_bn,explanation_en,reference`; BOM strip; 1,000-row cap; per-row validation; **all-or-nothing** — any error returns `{imported:0, failed:n, errors:[{row, messages}]}`, clean file imports in one transaction as drafts.
- **ModelTestService** — publish guard (≥ 1 question, all published); attach only published MCQs; reorder in transaction; **guardMutable**: structurally frozen once attempts exist (duplicate to edit).
- **TestAttemptService** — *start*: transaction + `lockForUpdate`, resume unexpired in-progress attempt, else create with server `expires_at`; *saveAnswer*: upsert, guarded by status/expiry/pivot membership; *submit*: lock, 30s grace, then *finalize* scores **from stored rows only** — correct + pivot marks, wrong -negative_mark, unanswered skipped; score rounded 2dp, may go negative; `active` → NULL. In-progress payloads never contain answers.
- **PracticeService** — published MCQs, random order, ≤ 50 , payload includes correct option + explanation (instant client-side feedback; login-only); `practice_sessions` record/history.
- **QuestionArchiveService** (public) — published MCQ (year-tagged) with answers inline, written list, server-built FilterDefs.
- **StudentService** (admin) — list withCount attempts, show stats, toggleActive.

A6. Permissions & seeders

`config/admin-permissions.php` += `view/create/edit/delete/publish questions`, `view/create/edit/delete/publish model tests`, `view students`, `edit students` (verbs auto-group in `RoleService`). `UserSeeder::STAFF_EXCLUDED` += `delete/publish questions`, `delete/publish model tests`, `edit students`. New additive `DemoExamPrepSeeder` (local only): ~20 Bar Council MCQs, 3 written questions, one published 10-question model test, demo student.

Part B — Frontend

B1. Student auth state

`HandleInertiaRequests::share()` adds a `student` prop (from the student guard) → typed in `inertia.d.ts`; new `useStudent()` hook (`{student, isLoggedInIn, logout}`). Server-side page gating via `auth:student` with locale-aware `?redirect=`; login page returns there after success.

B2. Student-facing pages (PublicLayout, locale-prefixed)

URL	Page
/exam-prep	Landing — practice / model test / archive cards + register CTA
/questions	Public archive — browse-page clone, server FilterDefs, mcq/written toggle, QuestionCard mode="archive"
/model-tests , /{slug}	Public catalogue + rules panel; Start (login-gated) / Resume
/model-tests/{slug}/attempts/{id}	Runner: rehydrates from server; countdown from server <code>remaining_seconds</code> (useCountdown hook); autosave (2s debounce + visibilitychange + 20s interval); QuestionPalette grid; submit confirm with unanswered count; auto-submit at expiry; beforeunload guard
/practice , /practice/run	Picker (program→subject cascade, stage/year, count 10/25/50) → one-question-at-a-time drill with instant reveal → summary (records practice_session)
/account/*	login, register, forgot/reset password, profile, attempts list, attempt result + per-question review

New shared pieces (only these): `components/public/QuestionCard.tsx` (modes archive/practice/runner/review — used by 5 screens), `QuestionPalette.tsx`, `hooks/useStudent.ts`, `hooks/useCountdown.ts`; `PublicHeader/MobileNavDrawer` get an Exam Prep item + login/account menu.

B3. Admin pages

- **pages/admin/questions/** — index (useResourceIndex, filters, status tabs, publish toggle), QuestionForm (type toggle hides options for written; 4–5 dynamic option rows with single correct radio; bilingual fields; `options.N.field` error mapping), create/edit, import (FileUpload + template download + per-row error table).
- **pages/admin/model-tests/** — index; create = settings only → redirects to edit; edit = settings + two-pane QuestionBuilder (filtered bank → add; ordered picks with up/down, per-question marks, running total; no drag library).
- **pages/admin/students/** — index with active/inactive tabs and activate/deactivate toggle (`edit students`).
- `sidebarNav.ts` : Questions / Model Tests / Students entries with permissions.

i18n namespaces added to `i18n.ts` (bn first): `nav.*`, `account.*`, `examprep.*`, `practice.*`, `mtest.*`, `qarchive.*`.

Part C — Tests

New Feature tests: `StudentAuthTest`, `StudentPasswordResetTest` (broker isolation), `ApiQuestionCrudTest` (exactly-one-correct rule, option sync, publish guard, 403s), `ApiQuestionImportTest` (all-or-nothing + row errors), `ApiModelTestBuilderTest` (frozen-after-attempt), `ApiStudentManagementTest`, `PublicQuestionArchiveTest`, `StudentPracticeTest`, `StudentModelTestAttemptTest` (resume, no answers in-progress, scoring 10q neg-0.25: 6c/2w/2s → 5.50, expiry, ownership 404). Update `PublicPageRouteTest` / `AdminSmokeTest`.

Implementation Order

1. Migrations + enums + models + `StudentResetPassword` notification
2. Auth plumbing (config/auth, bootstrap mount + guest redirect closure, limiter, `EnsureStudentIsActive`)
3. Student auth vertical + tests
4. Admin question bank + permissions + tests
5. CSV import + tests
6. Admin model tests + builder + tests
7. Student practice + attempt flow + tests
8. Public archive + public catalogue + tests
9. Admin student management + tests
10. Web page routes + page controllers + `DemoExamPrepSeeder`
11. Frontend: shared prop → account → archive → practice → runner → admin pages → nav
12. `vendor/bin/pint` + full `php artisan test` + `npx tsc --noEmit`

Locked-in Defaults (changeable later)

- Model tests freeze structurally after the first attempt — duplicate to edit (protects result integrity).
- Score not clamped at zero (Bar Council convention); 30-second grace on submit past expiry.
- CSV import all-or-nothing with a row-error report.
- No email verification and no per-attempt option shuffling in v1 (schema leaves room for both).
- Archive shows answers publicly by design (printed-guide analogy); model-test in-progress payloads never contain answers.